Language of Proof

Exam code: 9MA0

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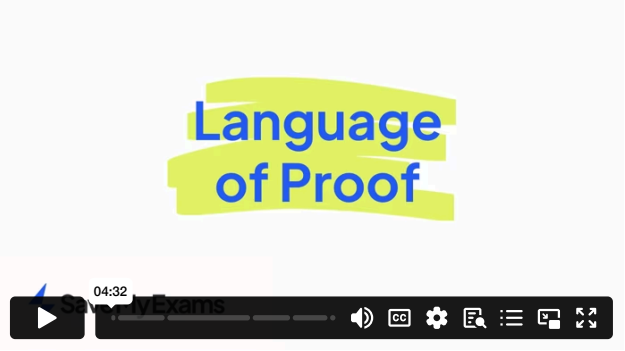

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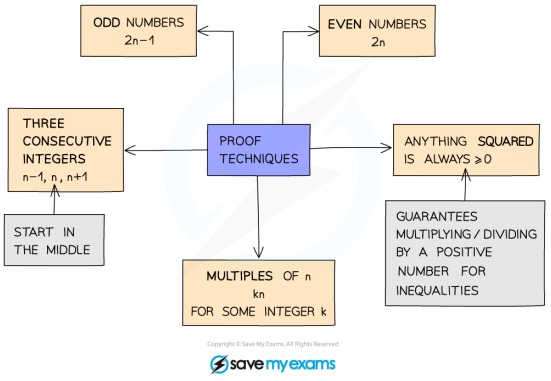
Help me understand

Language of proof

What is proof?

- Proof is a series of logical steps which show whether a result is **true** or not for a set of specified numbers
  - e.g. All integers, all even numbers etc

Notation in proof



- LHS and RHS are standard abbreviations for left-hand side and right-hand side
- Integers are used frequently in the language of proof

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$P=Q$	P IS EQUAL TO Q (TRUE FOR SOME VALUES)
$P \equiv Q$	P IS IDENTICAL TO Q (TRUE FOR ALL VALUES)
$P \Rightarrow Q$	P IMPLIES Q (IF P IS TRUE THEN Q IS TRUE)
$P \Leftrightarrow Q$	P IS EQUIVALENT TO Q (Q IS TRUE IF AND ONLY IF (IFF) P IS TRUE

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Worked Example

Prove that the sum of any three consecutive odd numbers is a multiple of 3.

LET THE THREE CONSECUTIVE ODD NUMBERS BE

$2n-3,\ 2n-1,\ 2n+1$

START WITH THE MIDDLE VALUE

THEN,

$(2n-3)+(2n-1)+(2n+1)$

$\equiv 6n-3$

$\equiv 3(2n-1)$

AIM FOR THE FORM 3k

WHICH IS A MULTIPLE OF 3

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